Import Dataset

In [36]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
```

In [37]:
```
df = pd.read_csv(r"D:\Downloads\Datasets\kaggle_survey_2017_2021.csv", low_memory=F
```

In [38]:
```
df
```

Out[38]:

	-	Time from Start to Finish (seconds)	Q1	Q2	Q3	Q4	Q5	Q6
0	Year	Duration (in seconds)	What is your age (# years)?	What is your gender? - Selected Choice	In which country do you currently reside?	What is the highest level of formal education ...	Select the title most similar to your current ...	For how many years have you been writing code ..
1	2021	910	50-54	Man	India	Bachelorâ€™s degree	Other	5-10 years
2	2021	784	50-54	Man	Indonesia	Masterâ€™s degree	Program/Project Manager	20+ years
3	2021	924	22-24	Man	Pakistan	Masterâ€™s degree	Software Engineer	1-3 years
4	2021	575	45-49	Man	Mexico	Doctoral degree	Research Scientist	20+ years
...	...	...	...	...	...	...	...	..
106297	2017	NaN	22-24	Female	Other	Master's degree	NaN	Less than a year
106298	2017	NaN	25-29	Male	Indonesia	Bachelor's degree	Programmer	1 to 2 years
106299	2017	NaN	25-29	Female	Taiwan	NaN	NaN	NaN
106300	2017	NaN	NaN	Female	Singapore	I prefer not to answer	NaN	NaN
106301	2017	NaN	25-29	Male	Japan	I did not complete any formal education past h...	Programmer	NaN

106302 rows × 293 columns

Explore the Data

In [57]:
```
#Temporarily Viewed:
```

```
#print(df.columns.tolist()) #See all column names in a list
#questions = df.iloc[0] # Grab the first row (the questions) befo


#for col, question in questions.items():
    #print(f"{col}: {question}")
```

Rename Columns

In [41]:
```
core_columns = {
    'Time from Start to Finish (seconds)': 'duration',
    'Q1': 'age',
    'Q2': 'gender',
    'Q3': 'country',
    'Q4': 'education',
    'Q5': 'job_title',
    'Q6': 'coding_experience',
    'Q8': 'recommended_language',
    'Q15': 'ml_experience',
    'Q20': 'industry',
    'Q25': 'salary',
    'Q41': 'primary_tool'
}

df.rename(columns=core_columns, inplace=True)
```

Keep Needed columns only

In [42]:
```
current_core_names = list(core_columns.values())
tool_cols = [col for col in df.columns if col.startswith(('Q7_Part', 'Q9_Part', 'Q1
# Reframe dataset

analysis_df = df[current_core_names + tool_cols].copy()
if analysis_df.iloc[0]['age'] == 'What is your age (# years)?':
    analysis_df = analysis_df.drop(0)

analysis_df.head()
```

Out[42]:

	duration	age	gender	country	education	job_title	coding_experience	rec
1	910	50-54	Man	India	Bachelorâ€™s degree	Other	5-10 years	
2	784	50-54	Man	Indonesia	Masterâ€™s degree	Program/Project Manager	20+ years	
3	924	22-24	Man	Pakistan	Masterâ€™s degree	Software Engineer	1-3 years	
4	575	45-49	Man	Mexico	Doctoral degree	Research Scientist	20+ years	
5	781	45-49	Man	India	Doctoral degree	Other	< 1 years	

5 rows × 45 columns

Clean the Data

In [43]:

```
analysis_df['education'] = analysis_df['education'].str.replace('â€™', "'", regex=F
analysis_df['job_title'] = analysis_df['job_title'].str.replace('â€™', "'", regex=F


for col in tool_cols:
    analysis_df[col] = analysis_df[col].notna().astype(int)                #Conver
```

Label Handling

In [44]:

```
le = LabelEncoder()
analysis_df['job_title_encoded'] = le.fit_transform(analysis_df['job_title'].astype


mapping = dict(zip(le.classes_, le.transform(le.classes_)))
print("Job Title Mapping:", mapping)           # Show the mapping for reference
```

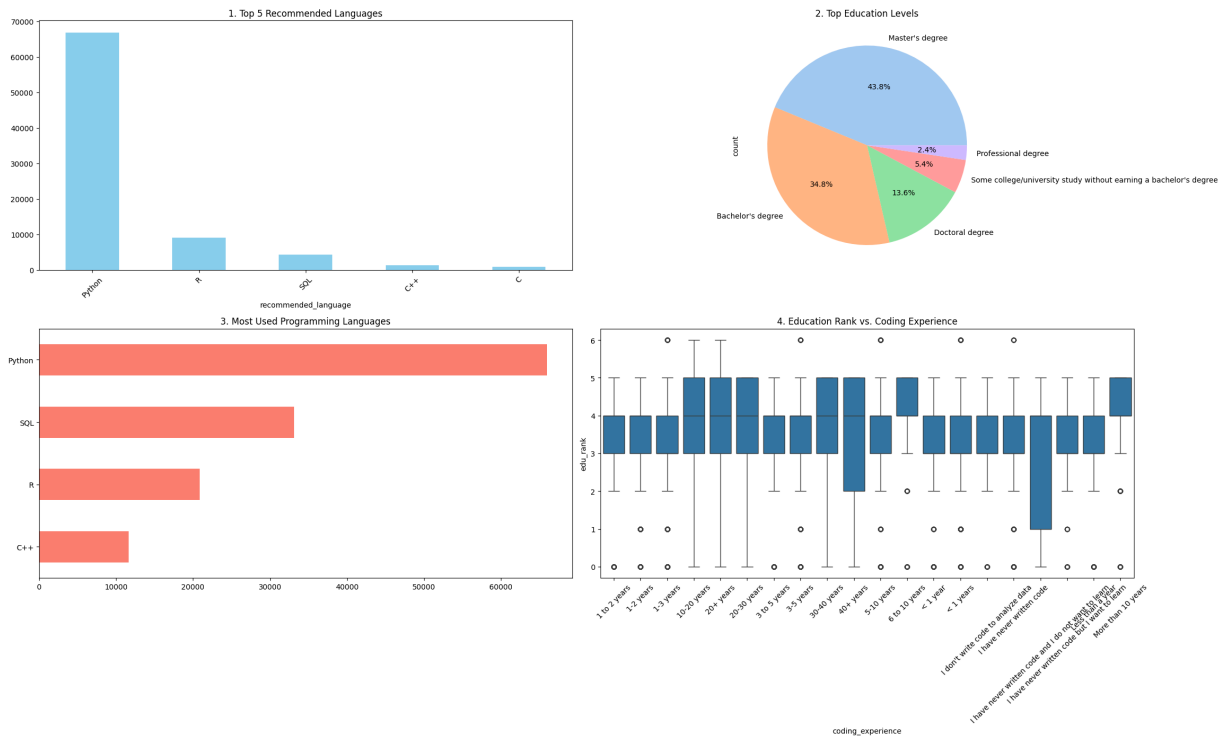
```
Job Title Mapping: {'Business Analyst': np.int64(0), 'Chief Officer': np.int64(1),
'Computer Scientist': np.int64(2), 'Consultant': np.int64(3), 'Currently not employe
d': np.int64(4), 'DBA/Database Engineer': np.int64(5), 'Data Analyst': np.int64(6),
'Data Engineer': np.int64(7), 'Data Journalist': np.int64(8), 'Data Miner': np.int64
(9), 'Data Scientist': np.int64(10), 'Developer Advocate': np.int64(11), 'Developer
Relations/Advocacy': np.int64(12), 'Engineer': np.int64(13), 'Machine Learning Engin
eer': np.int64(14), 'Manager': np.int64(15), 'Marketing Analyst': np.int64(16), 'Not
employed': np.int64(17), 'Operations Research Practitioner': np.int64(18), 'Other':
np.int64(19), 'Predictive Modeler': np.int64(20), 'Principal Investigator': np.int64
(21), 'Product Manager': np.int64(22), 'Product/Project Manager': np.int64(23), 'Pro
gram/Project Manager': np.int64(24), 'Programmer': np.int64(25), 'Research Assistan
t': np.int64(26), 'Research Scientist': np.int64(27), 'Researcher': np.int64(28), 'S
alesperson': np.int64(29), 'Scientist/Researcher': np.int64(30), 'Software Develope
r/Software Engineer': np.int64(31), 'Software Engineer': np.int64(32), 'Statisticia
n': np.int64(33), 'Student': np.int64(34), 'nan': np.int64(35)}
```

Insights

Explore the insights from Data

In [45]:
```
edu_map = {
    "No formal education past high school": 1,
    "Some college/university study without earning a bachelor's degree": 2,
    "Bachelor's degree": 3,
    "Master's degree": 4,
    "Doctoral degree": 5,
    "Professional doctorate": 6,
    "I prefer not to answer": 0
}
analysis_df['edu_rank'] = analysis_df['education'].map(edu_map).fillna(0)

# B. Label Encoding for Nominal Variables (Job Title & Gender)
le = LabelEncoder()
analysis_df['job_title_encoded'] = le.fit_transform(analysis_df['job_title'].astype
analysis_df['gender_encoded'] = le.fit_transform(analysis_df['gender'].astype(str))

print("Categorical variables have been encoded for ML.")
```

```
Categorical variables have been encoded for ML.
```

In [47]:
```
plt.figure(figsize=(24, 14))

#Top Recommended Languages
plt.subplot(2, 2, 1)
analysis_df['recommended_language'].value_counts().head(5).plot(kind='bar', color='
plt.title('1. Top 5 Recommended Languages')
plt.xticks(rotation=45)

#Education Level Distribution
plt.subplot(2, 2, 2)
analysis_df['education'].value_counts().head(5).plot(kind='pie', autopct='%1.1f%%',
plt.title('2. Top Education Levels')

#Language Usage (Binary tool columns)
plt.subplot(2, 2, 3)
```

```
lang_usage = {
    'Python': analysis_df['Q7_Part_1'].sum(),
    'R': analysis_df['Q7_Part_2'].sum(),
    'SQL': analysis_df['Q7_Part_3'].sum(),
    'C++': analysis_df['Q7_Part_5'].sum()
}
pd.Series(lang_usage).sort_values().plot(kind='barh', color='salmon')
plt.title('3. Most Used Programming Languages')

#Experience vs Education Rank
plt.subplot(2, 2, 4)
sns.boxplot(x='coding_experience', y='edu_rank', data=analysis_df.sort_values('codi
plt.title('4. Education Rank vs. Coding Experience')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```